2.1 Writing and Graphing Inequalities

Name _____ Period ____ Date _____

Algebra I 2.1 · Student sheet

I can…

- write inequalities from words.
- check solutions of inequalities and graph them on a number line.
- write inequalities from graphs and real-life situations.

Vocabulary · fill in the missing word

Inequality — a statement that compares two _____ with <, >, ≤, or ≥

Solution of an inequality — a value that makes the inequality _____

Graph of an inequality — open circle for < or >, closed circle for ≤ or ≥, with the _____ shaded

Part 1 · Words to symbols

Phrases: at least · at most · more than · fewer than · no more than · no less than · below · minimum · maximum · exceeds

<	>	≤	≥

Write one real-life sentence for each symbol.

Part 2 · Notes and examples (fill in each note, then work the example)

At least / at most: “At least 5” means x ____ 5. “At most 5” means x ____ 5.

Example 1

Write each sentence as an inequality.

a. A number x plus 4 is less than 10.

b. Twice a number y is no more than −6.

Solution of an inequality: A solution of an inequality is a value that makes it _____.

Example 2

Is x = −3 a solution of 2x + 5 ≥ −1?

Graphing: Open circle for _____ or _____; closed circle for _____ or _____.

Example 3

Graph each inequality.

a. $x > -2$ b. $y \leq 4$

Example 4

You must be at least 16 years old to get a driver’s license. Write and graph an inequality for the ages a that qualify.

Part 3 · Partner practice

Partner 1

Write as an inequality: a number t minus 3 is more than 8.

Partner 2

A number line has a closed circle at 1 and is shaded to the left. Write the inequality.